

Stepper Motor Specifications

Hybrid Stepper Motor NEMA 23 57HS, 2-Phase, 1.8°

General Characteristics

Item	Specification
Step angle accuracy	$\pm 5\%$
Resistance accuracy	$\pm 10\%$
Inductance accuracy	$\pm 20\%$
Temperature rise	80°C max
Ambient temperature	-20°C to +50°C
Insulation resistance	100 M Ω min. at 500 VDC
Dielectric strength	500 VAC for one minute
Radial shaft play	0.02 mm max. under 450 g load
Axial shaft play	0.08 mm max. under 450 g load

Electrical and Mechanical Specifications

Model	Step Angle	Rated Voltage	Rated Current	Phase Res.	Phase Ind.	Holding Torque	Detent Torque	Rotor Inertia	Motor Length	Leads	Weight
57HS112-3504	1.8°	2.45 V	3.5 A	0.7 Ω	3 mH	28 kg·cm	1200 g·cm	800 g·cm ²	112 mm	4	1.7 kg

Description

The NEMA 23 57HS112-3504 hybrid stepper motor is a two-phase stepper motor with a step angle of 1.8° . It is suitable for applications requiring accurate positioning, high holding torque, and reliable motion control, such as CNC machines, robotic systems, automation equipment, and 3D printing mechanisms.